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# The Interchangeability of the Cross-Platform Data in the Deep Learning-based Land Cover Classification Methodology

**Keisuke Yoshida[1], Shijun Pan[1], Satoshi Nishiyama[1], Takashi Kojima[2]**

[1] Graduate School of Environmental and Life Science, Okayama University
(Tsushima-naka 3-1-1, Kita-ku, Okayama 700-8530, Japan)
[2] TOKEN C. E. E. Consultants Co., Ltd

*Corresponding author: yoshida.k@okayama-u.ac.jp*

**Abstract.** This study employs a validated airborne Light Detection and Ranging (LiDAR) bathymetry system (ALB) and a UAV-borne Green LiDAR System (GLS) for cross-platform analysis of land cover classification (LCC). Furthermore, the study aims to visualize LiDAR data using high-contrast colour scales and improve the accuracy of land cover classification methods through image fusion techniques. If high-resolution aerial imagery is available, it needs to be downscaled to match the resolution of the low-resolution point clouds. The interchangeability of cross-platform data was assessed by comparing the interchangeability, which measures the absolute difference in overall accuracy (OA) or macro-F1 by comparing the cross-platform inter prediction. It is important to note that relying solely on aerial photographs is inadequate for achieving precise labelling, particularly under limited sunlight conditions that can result in misclassification. In such cases, LiDAR plays a crucial role in facilitating target recognition.

**Keywords:** Airborne LiDAR Bathymetry, Cross-platform, Deep Learning, Green LiDAR System, Riverine Land Cover Classification

## 1 Introduction

River environmental information includes crucially important data such as topographic bathymetry and vegetation attributes that are necessary to develop balanced river management measures, addressing issues such as flood control [1] and ecosystem management [2]. At past time, to get these mentioned data, field surveys that require personnel to entry to the site are necessary. These years, to overcome this limitation, digital surface models (DSMs) and digital terrain models (DTMs) have become indispensable tools for describing terrain conditions (i.e., using DSM minus DTM to infer the height of the land cover). Incidentally, Light Detection and Ranging (LiDAR) technology generates high-resolution and accurate DSMs and DTMs by halving the time between the emitted pulse and detection of the reflected echo [3]. Based on LiDAR technology characteristics, airborne LiDAR bathymetry (ALB) and unmanned aerial vehicle (UAV)-borne green LiDAR system (GLS) have been already applied in the riverine environment measurements in Japan [4]. In accordance with the mentioned measurements results (i.e., DSM minus DTM), how to classify the land cover using these results is becoming one of the tasks in the river engineering-related research.

These years, several approaches for overcoming land cover classification (LCC) difficulties have been tested, including decision tree algorithm [5], manual setting of thresholds [6], and deep learning [7]. Considering the time cost on discovering and extracting the feature of data, the deep learning method, particularly with the atrous convolution module has some advantage in effective feature finding. Therefore, DeepLabV3+ model with atrous convolution module was chosen for this study.

Subsequently, with the help of the DeepLabV3+ model, the high accuracy of LCC producing has been well proven from past study [7]. These studies utilized ortho-photographs with LiDAR data to classify the land cover using DeepLabV3+ model with additional module for adding LiDAR dataset with photographs. It also revealed that, because of the similar feature, recognition accuracy of LCC is high (i.e., averaged overall accuracy is almost 90%) when training and validation sets were selected from the similar location and period. However, this mentioned study was based on the same platform (e.g., training ALB dataset, predicting ALB dataset). If the data collected by different platforms, the impacts on the LCC mapping results derived by this operation has not been demonstrated yet. Thus, one more step, to mutual-predict cross-platform LCC to verify the interchangeability (i.e., train ALB dataset, predict GLS dataset; train GLS dataset, predict ALB dataset) becomes the target in this study. Alternatively, instead of using additional module for accuracy improvement, this study tried to increase the feature by image fusion derived from imagery and LiDAR. And how to transform the LiDAR data into imagery in an expressive way is also required in this study.

To observe the impact of cross-platform on LCC in this study as most as possible, the resolution and data styles (i.e., digital imagery, LiDAR) of ALB and GLS in mutual-prediction are chosen as same. On the other side, because of the data limitation, instead of totally same season, the similar season-related ALB and GLS dataset are selected.

Eventually, the comparisons in this research obtained from cross-platform mutual-predictions can facilitate understanding of cross-platform data features, and determine a reasonable method to retain the robustness using deep learning method with cross-platform data in LCC mapping producing.

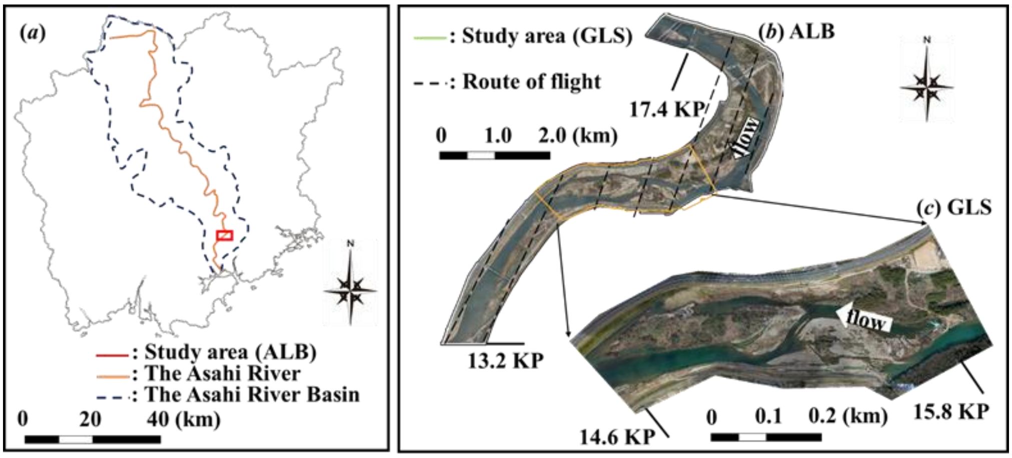


Figure 1 Perspective of Airborne LiDAR Bathymetry and Green LiDAR measurement area: (*a*) location of the Asahi River in Japan with kilo post (KP) values representing the longitudinal distance (km) from the river mouth, (*b*) aerial-captured photographs based on the marked positions in (*a*), and (*c*) drone-captured photographs based on the marked positions in (*b*).

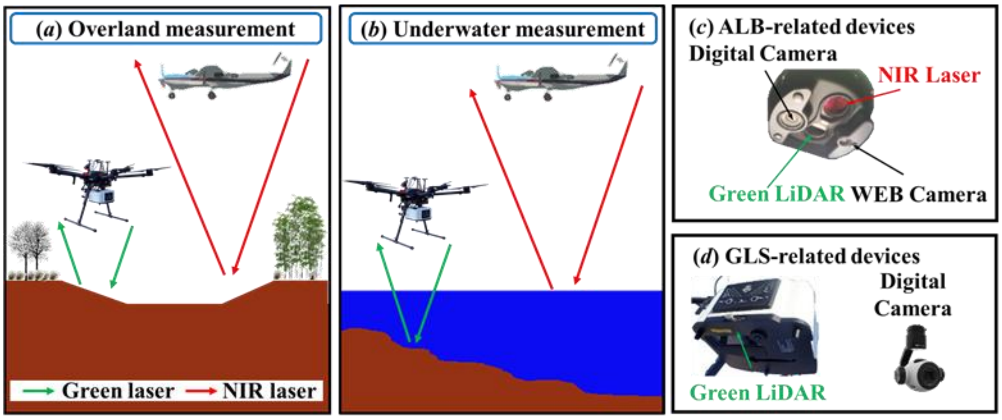


Figure 2 In (*a*) overland and (*b*) underwater surveys, Light Detection and Ranging (LiDAR) using a Near InfraRed (NIR) from (*c*) and green laser from (*d*), for ALB and GLS, respectively.

Table 1 Specifications of the present GLS and ALB system and measurement conditions in the targeted river reach.

| Items | | ALB | | GLS | |
|---|---|---|---|---|---|
| | | 2017 | | 2020 | |
| | | Mar. | Nov. | Mar. | Oct. |
| Laser wavelength range (nm) | Green | 515 | 515 | 532 | 532 |
| | NIR | 1,064 | 1,064 | - | - |
| Number of laser beams (103 s-1) | Green | 35 | 35 | 60 | 60 |
| | NIR | 148 | 148 | - | - |
| Ground altitude (m) | | 500 | 500 | 50 | 50 |
| Flight speed (km h-1) | | 220 | 110 | 9 | 9 |
| Density of measurement points (m-2) | Green | 2 | 4 | 100 | 100 |
| | NIR | 9.0 | 9.0 | - | - |
| Resolution of raw imagery (cm pixel-1) | | 10 | 10 | 3 | 3 |
| Resolution of raw LiDAR (m pixel-1)* | | 2 | 2 | 1 | 1 |
| FTU** | | - | - | 0.8 | 3.12 |
| NTU*** | | - | - | - | 3.7 |
| Degree**** | | 2.9 | 3.2 | - | - |

*: Based on the LiDAR-I.
**: Formazin Nephelometric Unit.
***: Nephelometric Turbidity Unit.
****: One degree of Japan Industrial Standard (JIS K0101) is the same as when 1 mg of standard substance (kaolin or formazine) is contained in 1 L of purified water.

# 2 Study site and methods

## 2.1 Study site

Figure 1 (a) depicts our study site, which is located on the lower Asahi River, a Class I (state-controlled) river in Japan, flowing through Okayama prefecture into the Seto Inland Sea. Throughout this study, the KP value denotes the longitudinal distance (kilometre, km) from the targeted river mouth. Besides, the riverbed slope is approximately 1:600. The channel width is about 300 m in the targeted reach. The targeted domain was 13.2–17.4 KP, as shown in Figure 1 (b), for the land cover classification (LCC).

As portrayed in Figure 1 (c), the GLS part was carried out in the middle of ALB part. The domain is a 1.2 km long counted from 14.6 KP to 15.8 KP, locally known as "Gion area". As a consequence of that, as shown in Figure 2 (a), (b), both of ALB and GLS measurement include overland and underwater area, and the land cover species all include the mentioned seven labels.

Appropriate to the flood control aspects, the riparian vegetation at this study site was classified roughly into three types based on flow resistance characteristic [7], which include bamboo forests (bamboo), herbaceous species (grass), and woody species (tree). Along with riverine vegetation, this study added another four labels (i.e., water, bare ground, road, and clutter) to represent local surface environmental changes better.

## 2.2 Data collection

### 2.2.1 ALB part

For this study, we conducted ALB (Leica Chiroptera II; Leica Corp.) surveys in March and November 2017 along a 4.2 km reach of the lower Asahi River (13.2–17.4 KP) controlled by the national government. As shown in Figure 1 (b), multiple flight operations were conducted in leaf-off (i.e., March 2017) condition to achieve overlapping coverage of the target area. The current system scanned the river channel for LCC using aircraft-mounted Near InfraRed (NIR) and green lasers as presented in Figure 2 (a). The device commonly uses the green laser to detect under-water (bottom) surfaces because green light can penetrate the water column to some degree. Conversely, the NIR laser is used to detect terrain surfaces, including vegetation, because it is readily reflected by the air-water interface. In the case of ALB, only NIR was used to calculate DSM and DTM. Moreover, during each ALB measurement, a digital camera as shown in Figure 2 (c), mounted directly beneath the aircraft took aerial photographs of the target river. Among other things, to remove tilt and relief effects, the aerial photographs were converted to orthophotos. Herein, the aerial photograph overlap and side-lap ratios were respectively greater than 60% and 30%. Table 1 shows specifications of the equipment, measurement parameters, and river water quality at the time of measurements. Because the magnitude of turbidity in a river can strongly affect the amount of light incident into the water column, its value was confirmed before each ALB measurement. The water quality of the three target periods was reasonable for measuring the underwater terrain surface.

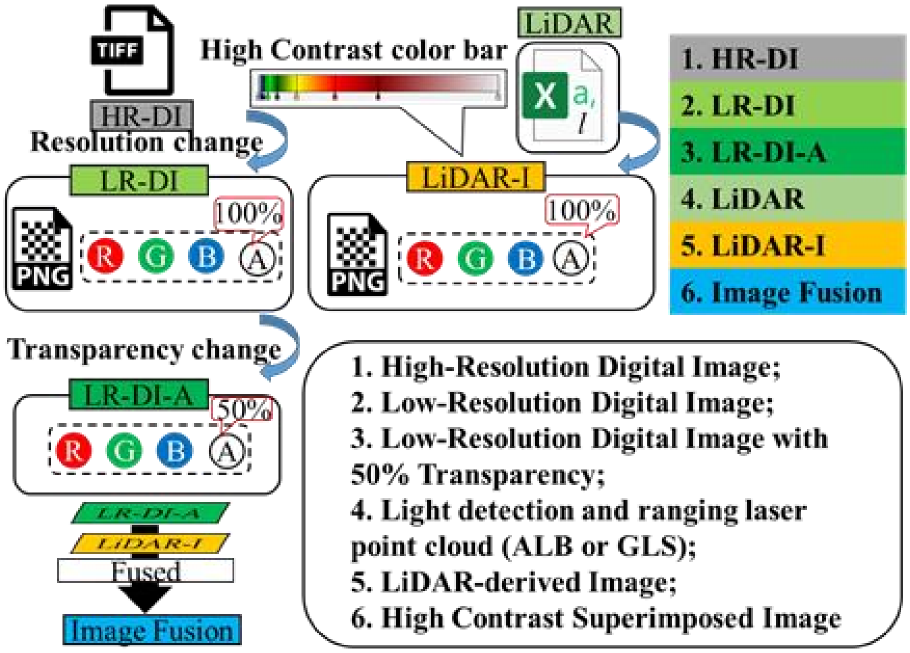


Figure 3 Preprocessing of imagery-based input data from drone images (aerial photographs) and LiDAR dataset (ALB or GLS).

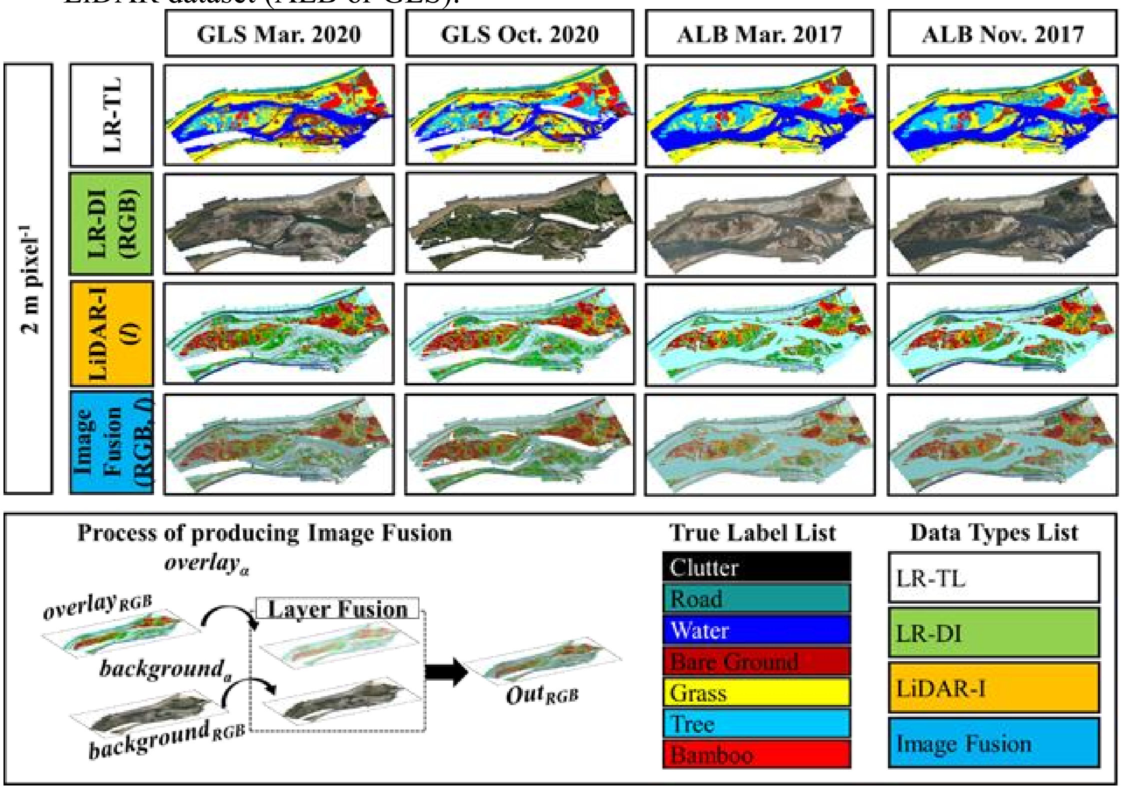


Figure 4 2 m pixel$^{-1}$ Imagery-based input (LR-TL, LR-DI, LiDAR-I, Image Fusion), Image Fusion processing, True Label List and Data Types.

### 2.2.2 GLS part

Both river topo-bathymetry and vegetation attribute measurements (i.e., LiDAR data, drone imagery) were performed on March and October 2020 with a normal water level utilizing digital drone-mounted GLS, respectively, through several flight operations. Figure 2 (b) illustrates a typical view of GLS during overland and underwater measurements. Table 1 shows specifications of the equipment in Figure 2 (d), measurement parameters, and river water quality at the time of measurements.

### 2.3 Data processing of imagery-based input

Figure 3 depicts the processing of producing imagery input data for this study [8]. It includes three types of data, i.e., LR-DI (low-resolution digital images derived from high-resolution digital images), LiDAR-I (images derived from LiDAR-based point cloud using a high-contrast color bar), and Image Fusion (images created by superimposing the above images).

#### 2.3.1 Pre-processing

In mapping the LCC pre-processing stage, we considered earlier field observation experience [7] to obtain critical assistance in mapping the true label (TL). Earlier field observation photographs can provide information about the study target LCC characteristics such as texture and color difference. Figure 4 presents an example of ALB- and GLS-based TL mapping (i.e., LR-TL).

#### 2.3.2 Transform high resolution to low resolution imagery (LR-DI)

The resolution conversion operation is necessary to ensure a consistent resolution of these input imagery. Incidentally, in order to test the cross-platform interchangeability of several data styles, the 2 m pixel$^{-1}$ resolution was chosen as shown in Figure 4. The resolution transformation in this study is utilizing a non-interpolation approach (i.e., a method that extracts only the color of pixels at fixed intervals) to keep the color of each pixel in the new generated image the same as that of the original image.

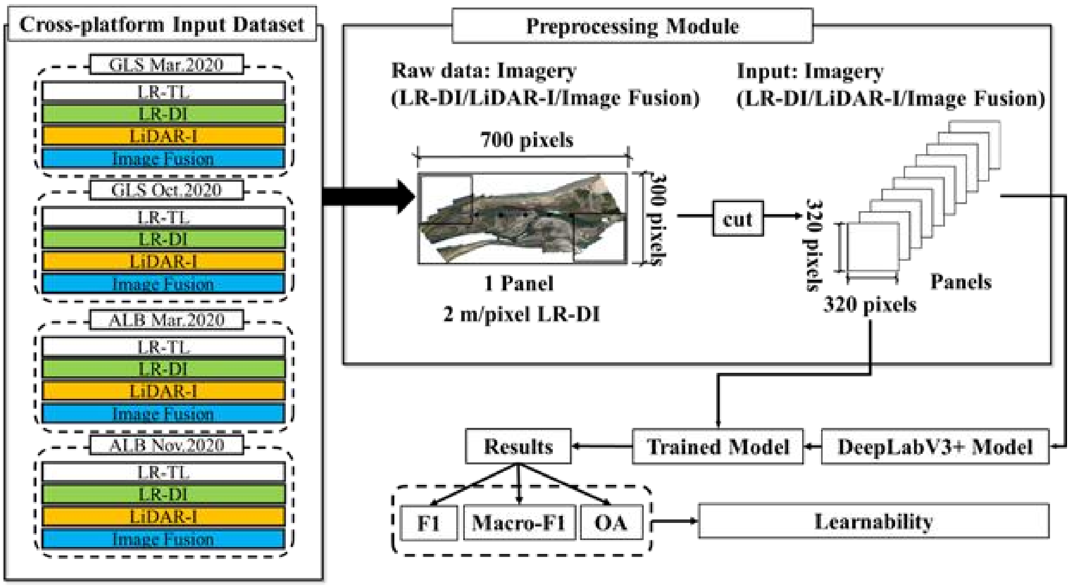


Figure 5 Processing of mapping LCC with DeepLabV3+ model; Overall accuracy (OA) is an accuracy measure that indicates how many of the total pixels are classified correctly; The macro-averaged F1 score (Macro-F1) is computed by taking the arithmetic mean (i.e., unweighted mean) of all the per-class F1 scores.

#### 2.3.3 Transform LiDAR data to imagery (LiDAR-I)

Raw GLS data includes laser point cloud coordinate, DSM minus DTM ($l$). To convert the $l$ information in csv file into visual imagery, the visualization software, Cloud Compare (i.e., 3-D point cloud and mesh processing software with open source project) was exploited. What's more, different visualization effects were achieved by changing the types of color scale or the distribution of the selected slider (i.e., the point to set specific color in the color scale).
Briefly, after comparing the visual effects of various default color scales, the "high-contrast color scale" was chosen for this study. Furthermore, the selected color scale displayed the LiDAR data as imagery after dividing the GLS data (from min to max) into 256 steps. Moreover, the "high-contrast

color scale" included six particular slides with different colors in the 0–50% segment of the color scale (i.e., 1%, 2%, 4%, 8%, 16%, and 32%).

This color scale was supportive for differentiating the laser point cloud better and for characterizing the data better by emphasizing even weakly distinguished image features. Because of the use of multiple split color bands, the "high-contrast color scale" provided better-distinguished data for the $l$ in the first half of the value. Subsequently, the visualized imagery was rendered to files using zoom-free method. Consequently, the output imagery maintained the same resolution of 1- (GLS) and 2- (ALB) m pixel$^{-1}$. Eventually, transform the 1- (GLS) to 2- (GLS) m pixel$^{-1}$ to keep same resolution.

### 2.3.4 Combine LR-DI with LiDAR-I (Image Fusion)

Because data collection using a single LiDAR sensor remain insufficient, the amount of information for imagery was augmented by integrating data from another sensor, such as a digital camera. Accordingly, image fusion approach, which is effective in the discipline of remote sensing, to create a fused image that includes clearer, more accurate, and more comprehensive information than from any single image. Actually, Image Fusion is a combination of two elements: LR-DI, LiDAR-I. Conclusively, Image Fusion was created and processed, as shown in Figure 4, using image processing software (i.e., GIMP) as the following steps: 1. Reduce the transparency of the overlay layer to 50%. 2. Maintain the original transparency of the background layer. 3. Merge these two layers into a single image with gamma correction (default method of layer combination in GIMP).

## 2.4 DeepLabV3+ Model

In the previous research [7], an approach of 3-channel DeepLabV3+ with an additional module to use LiDAR dataset as supplement was tried. And in this study, a 4-channel DeepLabV3+ model with modified input layer was also challenged for the higher accuracy. Nevertheless, because of the data limitation, it is very difficult for the 4-channel DeepLabV3+ model to extract the feature. Therefore, in this research, we just force on the effect of changing the input types and resolution of cross-platform dataset, without changing the internal of DeepLabV3+ model. As presented in Figure 5, to extract features from the input RGB-based imagery (i.e., LR-DI, LiDAR-I, Image Fusion), the processing of training and inference is represented as following: 1. Trimming the raw data from cross-platform input dataset with preprocessing module to attain imagery-based input data (320 pixel × 320 pixel), then training the input data with DeepLabV3+ model to achieve trained model; 2. Predicting the imagery-based input data with trained model. To end with attaining the results (i.e., OA and Mac-ro-F1 from mutual-prediction, separately); 3. Comparing the averaged and absolute difference value for confirming interchangeability of cross-platform dataset.

# 3 Results and discussion

Three data types of 2 m pixel$^{-1}$ input data were used to test the interchangeability as shown in Figure 5. The following four main comparison parameters, were utilized to quantify the interchangeability, i.e., Average (OA), Average (Macro-F1), Absolute Difference (OA) and Absolute Difference (Macro-F1), that were derived from 6 groups-based results in Table 2. Overall accuracy (OA) is an accuracy measure that indicates how many of the total pixels are classified correctly, subsequently, the macro-averaged F1 score (Macro-F1) is computed by taking the arithmetic mean (i.e., unweighted mean) of all the per-class F1 scores. Furthermore, lower Absolute Difference value means the stability of using cross-platform is much better. As shown in Figure 6, Image Fusion has improved in terms of the interchangeability rather than LR-DI and LiDAR-I. Training GLS Predicting ALB has much higher OA and Macro-F1 value using three data styles, separately. The reason of this phenomenon is the data unbalance derived from lack of the water area in GLS as shown in Figure 7. And without considering the water area, as shown in Table 2, OA and Macro-F1 derived from cross-platform become much similar rather than the results derived from Table 2.

Table 2 Comparison of 2 m pixel$^{-1}$ resolution cross-platform interchangeability using multiple method (LR-DI, LiDAR-I, Image Fusion) with/without considering the water area.

| Data styles | Groups | Train | | Predict | | Average (OA) | Average (Macro-F1) |
|---|---|---|---|---|---|---|---|
| | | GLS | ALB | GLS | ALB | | |
| LR-DI (RGB) | Group-1 | ○ | | | ○ | 0.73 (0.66) | 0.66 (0.66) |
| | Group-2 | | ○ | ○ | | 0.61 (0.62) | 0.59 (0.59) |
| LiDAR-I (l) | Group-3 | ○ | | | ○ | 0.72 (0.61) | 0.63 (0.63) |
| | Group-4 | | ○ | ○ | | 0.60 (0.64) | 0.56 (0.56) |
| Image Fusion (RGB, l) | Group-5 | ○ | | | ○ | 0.73 (0.63) | 0.65 (0.65) |
| | Group-6 | | ○ | ○ | | 0.64 (0.66) | 0.60 (0.60) |

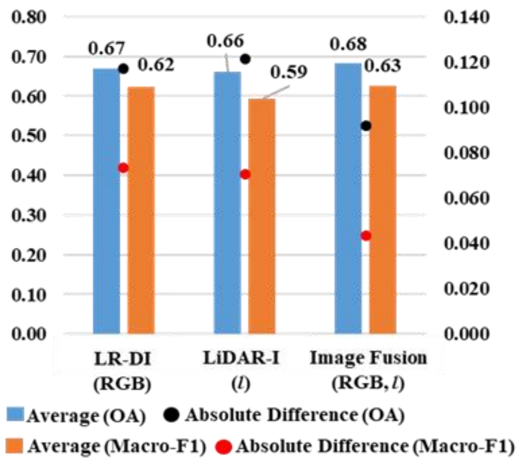


Figure 6 Comparison of data style-based averaged 2m pixel$^{-1}$ resolution cross-platform interchangeability; Left vertical axis: the reference of OA and Macro-F1 value; Right vertical axis: the reference of Absolute Difference value.

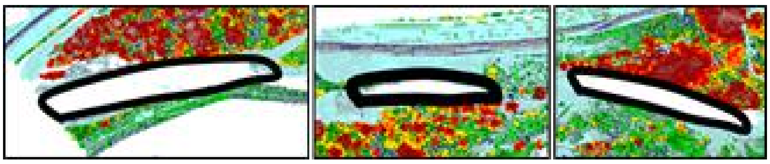

Figure 7 Water area that cannot be extracted with GLS only.

## 4 Conclusion

This study used the cross-platform LiDAR and photograph to prove the interchangeability be-tween the data derived from different platforms in the term of performing LCC. Compared with LR-DI and LiDAR-I, Image Fusion approach improved the performance of cross-platform LCC. And all the approaches have the results of over 0.65 OA and around 0.6 Macro-F1. To put it another way, to some content, cross-platform data can be used for inter-predicting each other. Noteworthy, digital imagery solely is not sufficient for producing TL mapping under multiple weather conditions because of the sensitive cam-era senor in the different weathers. Responsibly, at that moment, LiDAR becomes a considerable reference in assisting to recognize the targets.

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